complete list of physical characteristics they can attain is the same. Therefore, since the physical properties of the universe depend upon these properties of the basic constituents, there is no distinction, no way to differentiate, between radii that are inversely related to one another." Exactly.

Three Questions

At this point you might say, "Look, if I was a little being in the Garden-hose universe I would simply measure the circumference of the hose with a tape measure and thereby unambiguously determine the radius no ifs, ands, or buts. So what is this nonsense about two indistinguishable possibilities with different radii? Furthermore, doesn't string theory do away with sub-Planck distances, so why are we even talking about circular dimensions with radii that are a fraction of the Planck length? And finally, while we are at it, who really cares about the two-dimensional Garden-hose universe—what does all this add up to when we include all dimensions?"

Let's begin with the last question, as the answer will force us to come face to face with the first two.

Although our discussion has taken place in the Garden-hose universe, we restricted ourselves to one extended and one curled-up spatial dimension merely for simplicity. If we have three extended spatial dimensions and six circular dimensions—the latter being the simplest of all Calabi-Yau spaces—the conclusion is exactly the same. Each of the circles has a radius that, if interchanged with its reciprocal, yields a physically identical universe.

We can even take this conclusion one giant step further. In our universe, we observe three spatial dimensions, each of which, according to astronomical observations, appears to extend for about 15 billion light-years (a light-year is about 6 trillion miles, so this distance is about 90 billion trillion miles). As mentioned in Chapter 8, nothing tells us what happens after that. We do not know whether they continue on indefinitely or perhaps curve back on themselves in the shape of an enormous circle, beyond the visual sensitivity of state-of-the-art telescopes. If the latter is the case, an astronaut travelling out into space, continuously going in a fixed direction, would ultimately circle around the universe—like Magellan travelling around the earth—and wind up back at the initial starting point.